

# Geometry of Recursive Knots:

Iterated Cables by Frame Offsets, Their Thickness and Ropelength

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## Abstract

We study a concrete family of *recursive knots*: starting from a round circle  $K_0$ , the curve  $K_d$  is the  $(p_d, q_d)$  torus pattern placed on the tube of radius  $r_d$  around  $K_{d-1}$ , using a rotation-minimising (Bishop) frame closed by a linear twist. The question of the research line is how much rope (length divided by thickness) such nested knots need. We prove elementary facts for this family:  $K_d$  is an embedded smooth closed curve whenever  $r_d < \text{thick}(K_{d-1})$ ; the closed Bishop frame has linking number  $\text{Wr } K_{d-1} - \alpha_d/2\pi$  with its core, which identifies  $K_d$  as an explicit cable of  $K_{d-1}$ ;  $L(K_d)$  lies between  $p_d L(K_{d-1})(1 - r_d \kappa_{\max}(K_{d-1}))$  and  $p_d L(K_{d-1})(1 + r_d \kappa_{\max}(K_{d-1})) + r_d(2\pi q_d + p_d |\alpha_d|)$ ; and, for  $p_d = 2$ ,  $\text{thick}(K_d) \leq r_d$ , so that  $\text{Rop}(K_d) \geq L(K_d)/r_d$  inside the family. For the lower bound  $\text{thick}(K_d) \geq \min(\text{thick } K_{d-1} - r_d, c r_d \sin(\pi/p_d))$  we treat  $p_d = 2$ ,  $r_d \leq \text{thick}(K_{d-1})/2$ : in the default family (2,3) every doubly critical pair of  $K_d$  with distinct base points, on different strands or on the same strand (strands being defined relative to the shorter base arc), is at distance  $\geq \min(2(\text{thick } K_{d-1} - r_d), r_d)$ , at every depth and without further hypotheses (Lemma 3.7, Corollary 3.11; the same-strand case was obtained in the second round of review). The hypothesis  $(H_c)$  with  $c = 1/2$  therefore reduces to the curvature bound  $\min\text{Rad}(K_d) \geq r_d/2$ . That bound is proved at depth one, so  $\text{thick}(K_1) \geq r_1/2$  for every  $f = r_1 \leq 1/2$ ; for  $d \geq 2$  it follows from an explicit estimate under an additional bound  $(H_3)$  on the derivatives of the speed and of the parallel-frame curvature vector of  $K_{d-1}$ , whose constants we only evaluate on grids with inputs measured on the polygons (uncertified), and without such a bound no constant works for an arbitrary base (Remark 3.14). Under  $(H_c)$  the ropelength bound satisfies a linear recursion,  $\text{Rop}_d \leq 2\pi\Lambda^d$  with an explicit  $\Lambda$ . Numerically, explicit polygons with up to 8192 vertices give  $L/\tau = 38.4, 155.6$  and  $622.7$  at depths 1, 2, 3 for the default pattern (2,3) and offset fraction  $f = 1/2$ , stable to 0.03% under doubling of the resolution. These are numerically verified upper bounds for explicit polygons, not certified smooth thicknesses; no stationarity, optimality or recursive flow law is claimed.

## 1 Introduction and status

**The research line.** Line RTK001 (“Geometry of recursive knots”) asks how nested knots—knots tied on the tube of other knots, recursively—can be represented as tubes without self-intersection, and how much length that costs. Its public record states one finding (a feasible upper bound at depth one) and an explicit scope restriction: neither stationarity nor optimality of any configuration is certified, and no general recursive flow law has been established. No manuscript of the line existed before this draft; the family below was fixed from the record. Version 0.2 applied a first internal review (30/09/2026; the thickness hypothesis of v0.1 was false for an arbitrary base, Remark 3.14). Version 0.3 applies a second one (03/10/2026): every constant is labelled as analytic or as a grid evaluation, and the same-strand case, open in v0.2, is proved (Proposition 3.10), which proves  $(H_c)$  at depth one and reduces it to a curvature bound at higher depth (Theorem 3.13).

**What is done here.** We (i) fix a definite family of recursive knots (Section 2); (ii) prove embeddedness, the framing used, two-sided length bounds, an upper bound on the thickness and most of the lower bound, isolating what is not proved in one hypothesis on the family (Section 3); (iii) compute explicit polygonal configurations at depths 1 to 3, their polygonal thickness and ropelength, with a resolution check (Section 4); (iv) tabulate every claim with its status (Section 5). Lower bounds for general knots are only quoted from the literature (Remark 3.18); the only lower bound we derive is internal to the family (Proposition 3.5).

## 2 Setting

### 2.1 Thickness and ropelength

Let  $K \subset \mathbb{R}^3$  be a closed  $C^{1,1}$  curve of length  $L(K)$ . Its *thickness*  $\text{thick}(K)$  is the supremum of  $\rho$  such that the open tube of radius  $\rho$  about  $K$  is embedded (normal injectivity radius). Two equivalent descriptions are classical. Litherland, Simon, Durumeric and Rawdon [11] show, for  $C^2$  curves (extended to the  $C^{1,1}$  case in [3]; all curves  $K_d$  below are  $C^\infty$ ),

$$\text{thick}(K) = \min(\min\text{Rad}(K), \tfrac{1}{2} \text{dcsd}(K)), \quad (1)$$

where  $\min\text{Rad}$  is the minimal radius of curvature and  $\text{dcsd}$  the *doubly critical self distance*, the minimal distance between pairs  $x \neq y$  of  $K$  whose chord is perpendicular to  $K$  at both ends. Gonzalez and Maddocks [7] show  $\text{thick}(K) = \inf_{x \neq y} \rho_{pt}(x, y)$ , with  $\rho_{pt}(x, y) = |x - y| / (2 \sin \angle(T_x, y - x))$  the radius of the circle through  $y$  tangent to  $K$  at  $x$ . The *ropelength* is  $\text{Rop}(K) = L(K) / \text{thick}(K)$ ; it is scale invariant,  $\text{Rop}(K) \geq 2\pi$  for every closed curve (Fenchel's theorem [5], total curvature  $\geq 2\pi$ , together with  $\kappa \leq 1 / \text{thick}$ ), with equality only for the circle, and  $\text{thick}(K) \leq \min\text{Rad}(K)$  always.

### 2.2 Frames, holonomy and the closing twist

Let  $K$  be a closed  $C^2$  regular curve parametrised on  $[0, 2\pi)$  (not necessarily by arclength), with unit tangent  $T$  and curvature  $\kappa$ . A *rotation-minimising* or Bishop frame [1] is an orthonormal frame  $(T, N_0, B_0)$  with  $N'_0 = -\kappa_1 |K'| T$ ,  $B'_0 = -\kappa_2 |K'| T$ ,  $\kappa_1^2 + \kappa_2^2 = \kappa^2$ ; it is unique up to a constant rotation in the normal plane and is in general *not* periodic: after one turn the transported normal is rotated by a holonomy angle  $\alpha \in (-\pi, \pi]$ . We close it by the linear twist  $\omega(t) = -\alpha t / 2\pi$ , i.e.  $N = \cos \omega N_0 + \sin \omega B_0$ ,  $B = T \times N$ . The closed frame satisfies

$$N' = -\tilde{\kappa}_1 |K'| T + \omega' B, \quad B' = -\tilde{\kappa}_2 |K'| T - \omega' N, \quad |\omega'| = \frac{|\alpha|}{2\pi} \leq \frac{1}{2}, \quad (2)$$

with the rotated curvature components  $\tilde{\kappa}_1 = \kappa_1 \cos \omega + \kappa_2 \sin \omega$ ,  $\tilde{\kappa}_2 = -\kappa_1 \sin \omega + \kappa_2 \cos \omega$  (same norm  $\kappa$ ); below we write  $\kappa_1, \kappa_2$  for whichever frame is in use.

### 2.3 The family of recursive knots

**Definition 2.1** (iterated cable by frame offsets). Let  $K_0$  be the circle of radius  $R_0 = 1$ . For  $d \geq 1$  choose coprime integers  $p_d \geq 2$ ,  $q_d \geq 1$ , a radius  $r_d > 0$  and a phase  $\varphi_d$ . Reparametrise  $K_{d-1}$  to period  $2\pi$  by the linear change  $t \mapsto t/p_{d-1}$  (so that  $K_{d-1}$  is traversed at parameter speed  $|K'_{d-1}|$ , not at unit speed), let  $(N_{d-1}, B_{d-1})$  be its closed Bishop frame, and set for  $t \in [0, 2\pi p_d]$

$$K_d(t) = K_{d-1}(t \bmod 2\pi) + r_d [\cos \theta_d(t) N_{d-1}(t) + \sin \theta_d(t) B_{d-1}(t)], \quad \theta_d(t) = \frac{q_d}{p_d} t + \varphi_d. \quad (3)$$

The default is  $(p_d, q_d) = (2, 3)$ ,  $\varphi_d = 0$  at every depth;  $r_d$  is a fraction  $f$  of the thickness of  $K_{d-1}$  (the guaranteed lower bound  $\rho_{d-1}$  in the theory, the measured polygonal value in the experiments).

$K_1$  is the  $(p_1, q_1)$  torus knot on the round torus of radii  $(1, r_1)$ ; the knot type of  $K_d$  for  $d \geq 2$  depends on the framing, which Lemma 3.2 makes explicit. The pattern angle advances in the parameter  $t$ , not in arclength; since the parametrisation of  $K_{d-1}$  is the one inherited from the construction, the *pitch* of the strands relative to the base (the quantity  $h$  of Section 3) is a property of the family, not a free parameter.

### 2.4 Polygonal measurements

Numerically a curve is a closed polygon  $P = (x_0, \dots, x_{N-1})$ . We use  $L(P) = \sum |x_{i+1} - x_i|$ ;  $\min\text{Rad}(P)$ , the minimum circumradius of consecutive triples;  $\text{dcsd}(P)$ , the minimal distance  $|x_i - x_j|$  over *doubly critical vertex pairs*, i.e. pairs where  $|x_i - x_j|$  is a local minimum in  $i$  and in  $j$  (the diagonal  $D_{ii} = 0$  is kept in the neighbourhood test, which excludes near-diagonal pairs); and the proxy

$$\tau(P) = \min(\min\text{Rad}(P), \tfrac{1}{2} \text{dcsd}(P)), \quad \text{Rop}(P) = L(P) / \tau(P). \quad (4)$$

The local-minimum test could miss doubly critical pairs of saddle type; as a safeguard we also compute  $\tau_{pt}(P) = \min_{i \neq j} |x_j - x_i|^2 / (2|T_i \times (x_j - x_i)|)$  with central-difference tangents  $T_i$ , which uses no criticality

test. Neither  $\tau$  nor  $\tau_{pt}$  is a certified thickness of a smooth curve (for the thickness of polygons themselves see [13]); they are the quantities of (1) evaluated on an explicit polygon, and their stability under doubling of  $N$  is the main evidence offered that they approximate the smooth values. Vertex pairs may overestimate the polygon–polygon distance; the minimal segment–segment distance over pairs of edges with cyclic index separation  $> N/10$ , divided by the vertex dcscd, is 0.99994, 1.00000 ( $d = 1, f = 0.25, 0.5$ ) and 0.99993 ( $d = 2, f = 0.5$ ), and near the critical pair the ratio is  $\geq 0.99998$  in every level examined (`experiments/aux_checks.py`). Writhe is the Gauss double sum over pairs of edges (midpoint rule); its accuracy is discussed in Section 4.

### 3 Results

Throughout,  $K = K_{d-1}$  is a closed  $C^\infty$  regular curve of thickness  $\tau = \text{thick}(K) > 0$ , hence of maximal curvature  $\kappa_{\max} \leq 1/\tau$ , carrying the parametrisation of period  $2\pi$  inherited from Definition 2.1, with speed  $v = |K'| \in [v_{\min}, v_{\max}]$ ;  $K_d$  is built by (3) with  $r = r_d, p = p_d, q = q_d$ , and  $f = r/\tau$ .

**Lemma 3.1** (embedding). *[proved here] If  $r < \tau$  then  $K_d$  is a regular  $C^\infty$  closed curve, injective on  $[0, 2\pi p)$ , contained in the torus  $\partial\{x : \text{dist}(x, K) < r\}$ .*

*Proof.* Smoothness: the Bishop frame of a  $C^\infty$  curve is  $C^\infty$  (it solves a linear ODE with smooth coefficients), and (3) is a smooth expression in it; periodicity of  $K_d$  in  $t$  with period  $2\pi p$  holds because  $\theta_d(t + 2\pi p) = \theta_d(t) + 2\pi q$  and the closed frame has period  $2\pi$ . Regularity: differentiating (3) with (2),

$$K'_d = |K'| (1 - r(\kappa_1 \cos \theta + \kappa_2 \sin \theta)) T + r(\theta' + \omega')(-\sin \theta N + \cos \theta B), \quad (5)$$

and  $|\kappa_1 \cos \theta + \kappa_2 \sin \theta| \leq \kappa \leq 1/\tau$ , so the tangential coefficient is at least  $|K'|(1 - r/\tau) > 0$ . Injectivity: let  $t_1 \neq t_2$  in  $[0, 2\pi p)$ . If  $t_1 \not\equiv t_2 \pmod{2\pi}$ , the two points lie in the normal discs of radius  $r < \tau$  at distinct points of  $K$ ; by definition of thickness these discs are disjoint, so the points differ. If  $t_1 \equiv t_2 \pmod{2\pi}$ , then  $t_1 - t_2 = 2\pi k$  with  $0 < |k| < p$  and  $\theta_d(t_1) - \theta_d(t_2) = 2\pi qk/p \notin 2\pi\mathbb{Z}$  because  $\gcd(p, q) = 1$ ; the two points lie in the same normal disc at the same radius  $r > 0$  but at different angles, hence differ. Containment in the torus is immediate from (3).  $\square$

The constant  $1/2$  in  $r \leq \tau/2$ , used from Lemma 3.3 on, is what the estimates below need; embeddedness only needs  $r < \tau$ .

**Lemma 3.2** (the framing used, and the knot type). *[proved here, from classical results] Let  $\alpha \in (-\pi, \pi]$  be the holonomy of the Bishop frame of  $K$ . Then the closed frame  $(N, B)$  is a framing of  $K$  with linking number  $n := \text{Lk} = \text{Wr } K - \alpha/2\pi \in \mathbb{Z}$ ; in particular  $\alpha \equiv 2\pi \text{Wr } K \pmod{2\pi}$ , and  $n = \text{round}(\text{Wr } K)$  whenever  $\text{Wr } K \notin \mathbb{Z} + \frac{1}{2}$  (if  $\text{Wr } K \in \mathbb{Z} + \frac{1}{2}$  then  $\alpha = \pi$  and  $n = \text{Wr } K - \frac{1}{2}$ ). With the convention that a  $(p, q')$ -cable of  $K$  winds  $p$  times along  $K$  and  $q'$  times around it relative to a framing of linking number 0 (the Seifert framing),  $K_d$  is the  $(p, q + pn)$ -cable of  $K$  [10]. In particular  $K_1$  is the torus knot  $T(p_1, q_1)$ .*

*Proof.* By the Călugăreanu–White–Fuller formula [4, 6, 14],  $\text{Lk} = \text{Tw} + \text{Wr}$  for any framing. The Bishop frame has zero twist density; the closing rotation  $\omega$  adds total twist  $\text{Tw} = \frac{1}{2\pi} \int \omega' dt = -\alpha/2\pi \in [-\frac{1}{2}, \frac{1}{2})$ . Hence  $\text{Lk} = \text{Wr} - \alpha/2\pi$ , and since  $\text{Lk} \in \mathbb{Z}$  the congruence and the rounding rule follow. The pattern (3) winds  $q$  times around the meridian relative to the frame  $(N, B)$  while going  $p$  times along  $K$ ; relative to a framing of linking number 0 it winds  $q + pn$  times, which is the stated cable. For  $K_0$  a planar circle,  $\text{Wr} = 0 = \alpha$  and  $K_1$  is the standard torus knot.  $\square$

**Lemma 3.3** (length, two-sided). *[proved here] If  $r \leq \tau/2$  then*

$$p L(K) (1 - r\kappa_{\max}) \leq L(K_d) \leq p L(K) (1 + r\kappa_{\max}) + r(2\pi q + p|\alpha|) \leq p L(K) (1 + r/\tau) + r(2\pi q + \pi p). \quad (6)$$

*Proof.* From (5) and the triangle inequality,  $|K'_d| \leq |K'| (1 + r\kappa) + r(|\theta'| + |\omega'|) = |K'| (1 + r\kappa) + r(q/p + |\alpha|/2\pi)$ . Integrate over  $[0, 2\pi p)$ : the first term gives  $p \int_0^{2\pi} |K'| (1 + r\kappa) dt \leq p L(K) (1 + r\kappa_{\max})$ , the second  $r(2\pi q + p|\alpha|)$ . Then use  $\kappa_{\max} \leq 1/\tau$  and  $|\alpha| \leq \pi$ . For the lower bound,  $|K'_d|$  is at least its tangential component  $|K'| (1 - r\kappa) \geq |K'| (1 - r\kappa_{\max})$ ; integrate.  $\square$

*Remark 3.4.* The term  $p|\alpha|r \leq \pi pr$  from the closing twist, absent from a naive count “ $p$  copies plus  $q$  meridians”, is needed in the proof because the Bishop frame of a non-planar  $K$  does not close; numerically the bound without it also holds in all 36 levels (minimal ratio bound/ $L$  1.255).

**Proposition 3.5** (upper bound on the thickness and a lower bound on the ropelength,  $p = 2$ ). *[proved here] Let  $p = 2$  ( $q$  odd) and  $r < \tau$ . Then  $\text{thick}(K_d) \leq r$ , hence  $\text{Rop}(K_d) \geq L(K_d)/r \geq pL(K)(1 - r\kappa_{\max})/r$ . In the family with  $r_d = f \text{thick}(K_{d-1})$  and  $f \leq 1/2$  this gives  $\text{Rop}(K_d) \geq \frac{p(1-f)}{f} \text{Rop}(K_{d-1})$ , so  $\text{Rop}(K_d) \geq 2\pi(p(1-f)/f)^d$ .*

*Proof.* The two points  $K_d(t)$  and  $K_d(t + 2\pi)$  lie in the same normal disc of  $K$  at antipodal angles ( $\theta$  and  $\theta + \pi q$ ,  $q$  odd), so their chord is  $-2r(\cos \theta N + \sin \theta B)$ , orthogonal to  $T$  and to  $-\sin \theta N + \cos \theta B$ , hence by (5) orthogonal to  $K'_d$  at both ends: the pair is doubly critical at distance  $2r$ , so  $\text{dcsd}(K_d) \leq 2r$  and  $\text{thick}(K_d) \leq r$  by (1) (equivalently,  $\rho_{pt} = r$  for this pair). The rest is Lemma 3.3 with  $\kappa_{\max} \leq 1/\tau$ ,  $r = f\tau$  and  $\text{Rop}(K_{d-1}) = L(K_{d-1})/\tau$ .  $\square$

For the default family at  $f = 1/2$  the internal lower bound is  $2\pi \cdot 2^d$ , i.e. 12.6, 25.1, 50.3 at  $d = 1, 2, 3$ , against the measured 38.4, 155.6, 622.7; with the measured  $L(K_d)$  the bound  $L(K_d)/r_d$  is attained whenever  $\text{thick}(K_d) = r_d$  (Section 4). The lower bound of (6) holds in all 36 measured levels (36 of 36).

**Thickness of  $K_d$ : what is proved and what is not.** Write  $s = 2r \sin(\pi/p)$  for the chord between adjacent strands in a normal disc of  $K$  (the  $p$  points  $K_d(t + 2\pi k)$ ,  $k = 0, \dots, p-1$ , sit at the  $p$  equally spaced angles  $\theta + 2\pi qk/p$ , since  $\gcd(p, q) = 1$ ).

**Proposition 3.6** (two of the three cases). *[proved here] If  $r \leq \tau/2$ : (a) two points of  $K_d$  in the same normal disc of  $K$  are at distance  $\geq s$ ; (b) two points of  $K_d$  whose base points on  $K$  are at distance  $\geq 2\tau$  are at distance  $\geq 2(\tau - r)$ .*

*Proof.* (a) They are at the same radius  $r$  and at angles differing by  $2\pi qk/p$ ,  $0 < k < p$ , so their distance is  $2r|\sin(\pi qk/p)| \geq 2r \sin(\pi/p) = s$  because  $qk \not\equiv 0 \pmod{p}$ . (b) Triangle inequality:  $|K_d(t_1) - K_d(t_2)| \geq |K(t_1) - K(t_2)| - 2r$ .  $\square$

Not covered by Proposition 3.6 are the pairs with  $0 < |K(t_1) - K(t_2)| < 2\tau$ , and the curvature of  $K_d$ . These include nearby base points (the helix geometry of  $p$  strands on a bent tube) and base points on different arcs of  $K$ , e.g. across the hole of a torus: at  $f = 1/2$ ,  $d = 1$  the doubly critical vertex pair that realises  $\text{dcsd}$  has base points  $103.7^\circ$  apart on  $K_0$ , at distance  $1.573 < 2\tau_0 = 2$  (`aux_checks.py`), while for  $f \leq 0.35$  it is the antipodal pair of one normal disc. For a straight tube the  $p$  strands are helices with pitch parameter  $h = |K'|p/q$  and the antipodal pair ( $p = 2$ ) is a local minimum of the strand distance exactly when  $h \geq r$ ; at depth one the relevant pitch on the inner equator is  $(1-r)p/q$ , and  $(1-r)p/q \geq r$  iff  $r \leq p/(p+q) = 0.4$  for  $(2, 3)$ , consistent with  $\text{thick}(K_1) = r_1$  for  $f \leq 0.35$  and  $\text{thick}(K_1) < r_1$  for  $f = 0.5$ .

**Notation for the local case.** Let  $(T, N_0, B_0)$  be the (non-periodic) Bishop frame of  $K$  and  $\alpha$  its holonomy, and set

$$m := \frac{q}{p} - \frac{\alpha}{2\pi}, \quad h_{\min} := \frac{v_{\min}}{m}, \quad \Theta_{\min} := \frac{h_{\min}}{\tau}, \quad \zeta_{\min} := \frac{(1-f)h_{\min}}{r}, \quad f = \frac{r}{\tau} \leq \frac{1}{2}.$$

In the parallel frame the strand of (3) turns at the constant rate  $m$  (its angle is  $\psi = \theta + \omega = mt + \varphi$ ), so  $h = v/m$  is the pitch parameter above ( $|K'|p/q$  when  $\alpha = 0$ ) and  $\Theta_{\min}$  is the smallest pitch in units of the base thickness. For two points  $X_i = K(t_i) + rU_i$  of  $K_d$  with distinct base points, let  $\sigma$  be the length of the *shorter* base arc between  $K(t_1)$  and  $K(t_2)$ ,  $\delta = \sigma/\tau$ ,  $s$  the base-parameter difference along that arc and  $u = m|s|$  the relative turning of the strands along it. The integer  $k = (t_2 - t_1 - s)/2\pi \pmod{p}$  is the strand index of  $X_2$  relative to  $X_1$  *along that arc* (for  $p = 2$  the two strands are exchanged by one turn around  $K$ , so “same strand” is meaningful only relative to a base arc). With  $\ell_0(\delta) = 2\tau \sin(\delta/2)$ ,  $E(\delta) = \tau(1 - \cos \delta)$  for  $\delta \leq \pi/2$  and  $E(\delta) = \tau(1 + \delta - \pi/2)$  for  $\delta > \pi/2$ ,  $\beta(\delta) = \delta^2/(4 \sin(\delta/2))$ , define

$$\Lambda(\delta) = \ell_0 - \frac{2rE}{\ell_0}, \quad P_1(\delta) = \left[ 2 \cos\left(\frac{\delta}{2\Theta_{\min}}\right) \sqrt{1 - \beta(\delta)^2} - \frac{\delta^2}{3} \right]_+ \quad (7)$$

(with  $P_1 := 0$  if  $\delta > \pi\Theta_{\min}$  or  $\beta > 1$ ), and

$$c_1(f, \Theta_{\min}) := \frac{1}{2r} \inf_{0 < \delta < \pi} \max \left\{ \sqrt{\Lambda(\delta)_+^2 + r^2 P_1(\delta)^2}, \ell_0(\delta) - 2r \right\}. \quad (8)$$

All lengths in (8) scale with  $r$ , so  $c_1$  depends only on  $(f, \Theta_{\min})$ ;  $\ell_0/r - 2$  and  $\Lambda/r = (2/f) \sin(\delta/2) - (E/\tau)/\sin(\delta/2)$  decrease in  $f$  and  $P_1$  does not depend on  $f$  and increases in  $\Theta_{\min}$ , so  $c_1$  is non-increasing in  $f$  and non-decreasing in  $\Theta_{\min}$ .

**Lemma 3.7** (pairs on different strands,  $p = 2$ ). *[proved here] Let  $p = 2$ ,  $q$  odd,  $r \leq \tau/2$ . Any two points  $X_1, X_2$  of  $K_d$  with distinct base points and strand index  $k = 1$  relative to the shorter base arc satisfy*

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c_1(f, \Theta_{\min})r).$$

Moreover  $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$  (a grid evaluation gives 0.6117), hence  $c_1 \geq \frac{1}{2}$  whenever  $f \leq \frac{1}{2}$  and  $\Theta_{\min} \geq \frac{2}{3}$ . For the default family  $(p, q) = (2, 3)$  with  $r_j \leq \text{thick}(K_{j-1})/2$  for all  $j \leq d$  one has  $\Theta_{\min} \geq \frac{2}{3}$  at every depth, so every such pair is at distance  $\geq \min(2(\tau - r), r)$ .

*Proof. Step 1 (chord of a thick curve).* Let  $x = K(0)$  and  $g(\sigma) = |K(\sigma) - x|$  in arclength. A local minimum  $y$  of  $g$  in  $(0, L)$  has  $(y - x) \perp T(y)$ , hence  $\rho_{pt}(y, x) = |x - y|/2 \geq \tau$ , so  $|x - y| \geq 2\tau$ . Consequently  $\{g < 2\tau\}$  consists of an initial interval  $[0, a)$  and a final one  $(L - b, L)$  on which  $g$  is monotone (a non-monotone stretch below  $2\tau$  would contain an interior local minimum below  $2\tau$ ), and on them  $g' = \cos \angle(T, K - x) \geq \sqrt{1 - g^2/4\tau^2}$  a.e. (from  $\rho_{pt}(K(\sigma), x) \geq \tau$ ), which integrates to  $g(\sigma) \geq 2\tau \sin(\sigma/2\tau)$  and shows  $a, b \leq \pi\tau$  (this is the classical chord-arc inequality for thick curves, cf. [3, 7]). Hence for base points with shorter arc  $\sigma = \tau\delta \leq L/2$ : either  $\delta \geq \pi$  and  $|x - y| \geq 2\tau$  (then Proposition 3.6(b) gives  $|X_2 - X_1| \geq 2(\tau - r)$ ), or  $\delta < \pi$  and  $\ell := |x - y| \geq \ell_0(\delta)$ .

*Step 2 (elementary estimates,  $\delta < \pi$ ).* Let  $y = K(t_2)$ ,  $e = (y - x)/\ell$  and  $\bar{T} = \sigma^{-1} \int vT dt$  over the arc, so  $e = \bar{T}/|\bar{T}|$ ,  $|\bar{T}| = \ell/\sigma$ . Since  $|dT/d\sigma| = \kappa \leq 1/\tau$ : (E2)  $|\bar{T} - T(x)| \leq \delta/2$ ; (E3) for  $w \perp T(x)$ ,  $|w \cdot e| = |w \cdot \Pi_x e| \leq |w| |\bar{T} - T(x)| \sigma/\ell \leq |w| \beta(\delta)$ , where  $\Pi_x$  is the projection onto the normal plane at  $x$ ; (E4) for unit  $w \perp T(x)$ ,  $|w \cdot (y - x)| = |\int v w \cdot \Pi_x T dt| \leq \int_0^\sigma \sin \min(\sigma'/\tau, \pi/2) d\sigma' = E(\delta)$ , and likewise for  $w \perp T(y)$ ; (E5) if  $\tilde{U}_2$  denotes the vector with the same parallel-frame angle as  $U_2$  but in the normal plane at  $x$ , then  $U_2 - \tilde{U}_2 = -\int \kappa_\perp vT dt$  with  $|\kappa_\perp| \leq \kappa$  (differentiate  $\cos \psi_2 N_0 + \sin \psi_2 B_0$  along the arc), so  $|U_2 - \tilde{U}_2| \leq \delta$  and its component orthogonal to  $e$  is at most  $\int \kappa v |T - \bar{T}| dt \leq \int_0^\sigma \frac{\sigma'^2 + (\sigma - \sigma')^2}{2\sigma\tau^2} d\sigma' = \delta^2/3$ ; (E6)  $\tilde{U}_2 - U_1$  lies in the normal plane at  $x$  and has length  $2|\sin(\Delta\psi/2)|$ , where  $\Delta\psi = ms + 2\pi qk/p$  is the relative angle of the strands after parallel transport along the arc (the holonomy terms cancel because  $N_0(t + 2\pi)$  is  $N_0(t)$  rotated by  $\alpha$  and  $m$  contains  $-\alpha/2\pi$ ), i.e.  $|\sin(\Delta\psi/2)| = |\cos(u/2)|$  for  $k = 1$  and  $p = 2$ .

*Step 3 (decomposition).* Write  $X_2 - X_1 = (y - x) + r(U_2 - U_1) = Ae + Q$  with  $Q \perp e$ . By (E4),  $A = \ell + r(U_2 - U_1) \cdot e \geq \ell - 2rE/\ell \geq \Lambda(\delta)$ , since  $\ell \mapsto \ell - 2rE/\ell$  is increasing and  $\ell \geq \ell_0$ . By (E3), (E5), (E6),  $|Q| = r|(U_2 - U_1)_\perp| \geq r[2|\cos(u/2)|\sqrt{1 - \beta^2} - \delta^2/3]_+$ . Finally  $u = m|s| \leq m\sigma/v_{\min} = \delta/\Theta_{\min}$  and  $|\cos(u/2)|$  is decreasing on  $[0, \pi]$ , so  $|\cos(u/2)| \geq \cos(\delta/2\Theta_{\min})$  when  $\delta \leq \pi\Theta_{\min}$ ; together with the trivial bound  $|X_2 - X_1| \geq \ell - 2r \geq \ell_0 - 2r$  this gives  $|X_2 - X_1| \geq 2c_1(f, \Theta_{\min})r$  for every pair with  $\delta < \pi$ .

*Step 4 ( $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ ).* Put  $r = 1$ ,  $\tau = 2$ ,  $\Theta_{\min} = 2/3$ ; we show that the maximum in (8) is  $\geq 1$  for every  $\delta \in (0, \pi)$ . For  $\delta \in [\pi/3, \pi/2]$ ,  $\Lambda = 2(\tau - r)\sin(\delta/2) = 2\sin(\delta/2) \geq 1$ . For  $\delta \in [\pi/2, \pi]$ , with  $\varsigma = \sin(\delta/2) \in [1/\sqrt{2}, 1]$ ,  $\Lambda = 4\varsigma - (1 + 2\arcsin \varsigma - \pi/2)/\varsigma$ , and  $\Lambda \geq 1$  amounts to  $\phi(\varsigma) := 4\varsigma^2 - \varsigma - 1 + \pi/2 - 2\arcsin \varsigma \geq 0$ ;  $\phi'' = 8 - 2\varsigma(1 - \varsigma^2)^{-3/2}$  changes sign once on the interval, so  $\phi'$  is increasing then decreasing, and since  $\phi'(1/\sqrt{2}) > 0$  and  $\phi'(1^-) = -\infty$ ,  $\phi$  increases then decreases; its minimum is at an endpoint,  $\phi(1/\sqrt{2}) = 0.29$ ,  $\phi(1) = 0.43$ . For  $\delta \in (0, \pi/3)$  we need  $4\sin^2(\delta/2) + P_1^2 \geq 1$ . On  $[0.6, \pi/3]$ :  $4\sin^2(\delta/2) \geq 4\sin^2(0.3) > 0.349$ ,  $\cos(3\delta/4) \geq \cos(\pi/4) > 0.7071$ ,  $\beta \leq \beta(\pi/3) < 0.5484$  ( $\beta$  is increasing), so  $\sqrt{1 - \beta^2} > 0.8362$ , and  $\delta^2/3 \leq 0.3656$ ; hence  $P_1 > 2 \cdot 0.7071 \cdot 0.8362 - 0.3656 > 0.8169$ ,  $P_1^2 > 0.667$ , and the sum exceeds 1.01. On  $(0, 0.6]$ :  $\cos(3\delta/4) \geq \cos(0.45) \geq 0.900$ ,  $\beta \leq \beta(0.6) \leq 0.305$ ,  $\delta^2/3 \leq 0.12$ , so  $P_1 \geq 2 \cdot 0.900 \cdot 0.952 - 0.12 \geq 1.59$ . All inequalities are non-strict where needed: at  $\delta = \pi/3$ ,  $\Lambda = 1$  exactly. Hence  $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ . The value 0.6117 (attained near  $\delta = 1.23$ ; an independent evaluation in the review gave 0.611895) is a grid infimum with a Lipschitz constant estimated on the grid itself (experiments/check\_Hc.py), not a certified bound.

*Step 5 (the family).*  $\Theta_{\min} = v_{\min}/(m\tau)$ . By Proposition 3.5,  $\tau_{d-1} = \text{thick}(K_{d-1}) \leq r_{d-1} \leq \tau_{d-2}/2$ , so  $\tau_{d-1} \leq 2^{-(d-1)}$ . After the reparametrisation  $t \mapsto t/p$ , (5) gives  $v_{d,\min} \geq p(1 - f)v_{d-1,\min} \geq v_{d-1,\min}$  for  $p = 2$ ,  $f \leq 1/2$ , so  $v_{\min} \geq v_0 = 1$  at every depth; and  $m \leq q/p + 1/2 = 2$ . Hence  $\Theta_{\min} \geq 2^{d-2} \geq 1$  for  $d \geq 2$ , while for  $d = 1$  ( $\alpha = 0$ ,  $m = 3/2$ ,  $v = \tau = 1$ )  $\Theta_{\min} = 2/3$ . Monotonicity of  $c_1$  concludes.  $\square$

**Proposition 3.8** (curvature of  $K_d$ ). *[proved here, under (H<sub>3</sub>)] Assume in addition (H<sub>3</sub>):  $\sup |v'| \leq M_v$  and  $\sup \sqrt{\kappa_1'^2 + \kappa_2'^2} \leq M_\kappa$  for  $K$  (derivatives in  $t$ , parallel-frame curvatures;  $\kappa_1'^2 + \kappa_2'^2 = \kappa'^2 + v^2\kappa^2\tau_{\text{tors}}^2$  includes the torsion). Then, with  $g(\zeta) = \zeta(\zeta^2 + 2)/(\zeta^2 + 1)^{3/2}$  and  $G(\zeta_{\min}) = \sup_{\zeta \geq \zeta_{\min}} g \leq g(\sqrt{2}) = 1.0887$ ,*

$$\kappa_{K_d} \leq \bar{\kappa}_d := \frac{G(\zeta_{\min})}{\tau - r} + \frac{1}{r(1 + \zeta_{\min}^2)} + \frac{rm[M_v(1 + f) + r v_{\max} M_\kappa]}{(v_{\min}^2(1 - f)^2 + r^2 m^2)^{3/2}}, \quad \text{minRad}(K_d) \geq 1/\bar{\kappa}_d. \quad (9)$$

*Proof.* Let  $U = \cos \psi N_0 + \sin \psi B_0$ ,  $W = T \times U$ ,  $\kappa_\perp = \kappa_1 \cos \psi + \kappa_2 \sin \psi$ ,  $\kappa_W = -\kappa_1 \sin \psi + \kappa_2 \cos \psi$  ( $\kappa_\perp^2 + \kappa_W^2 = \kappa^2$ ),  $x = r\kappa_\perp \in [-f, f]$ . Differentiating (5) with  $U' = mW - \kappa_\perp vT$ ,  $W' = -mU - \kappa_W vT$ ,  $T' = v(\kappa_\perp U + \kappa_W W)$ :

$$\begin{aligned} K_d' &= c_T T + c_W W, & K_d'' &= aT + b_U U + b_W W, \\ c_T &= v(1-x), & c_W &= rm, & b_U &= v^2(1-x)\kappa_\perp - rm^2, & b_W &= v^2(1-x)\kappa_W, \end{aligned}$$

$a = a_0 - 2rv m \kappa_W$ ,  $a_0 = v'(1-x) - rv(\kappa_1' \cos \psi + \kappa_2' \sin \psi)$ . Since  $(T, U, W)$  is orthonormal and right-handed,  $|K_d' \times K_d''|^2 = b_U^2 |K_d'|^2 + (c_W a - c_T b_W)^2$ . Put  $A = v^2(1-x)^2$ ,  $B = r^2 m^2$ , so  $|K_d'|^2 = A + B$  and  $c_W a - c_T b_W = r m a_0 - \kappa_W v(A + 2B)$ . By Minkowski,  $\kappa_{K_d} \leq \sqrt{v^4(1-x)^2 \kappa_\perp^2 (A+B) + \kappa_W^2 v^2 (A+2B)^2} / (A+B)^{3/2} + (rm^2(A+B)^{1/2} + rm|a_0|) / (A+B)^{3/2}$ . In the first term  $A(A+B) \leq (A+2B)^2$  and  $\kappa_\perp^2 + \kappa_W^2 = \kappa^2$  give  $\leq v\kappa(A+2B)/(A+B)^{3/2}$ , which is decreasing in  $A \geq v^2(1-f)^2$ ; substituting  $A = v^2(1-f)^2$  and  $\zeta = v(1-f)/(rm)$  yields  $\kappa g(\zeta)/(1-f) \leq G(\zeta_{\min})/(\tau(1-f))$ . The second term is at most  $1/(r(1+\zeta_{\min}^2)) + rm[M_v(1+f) + rv_{\max} M_\kappa]/(v_{\min}^2(1-f)^2 + r^2 m^2)^{3/2}$ .  $\square$

**Same-strand pairs by a turning argument (added in v0.3).** For a pair on the same strand ( $k = 0$ ) Step 3 of Lemma 3.7 gives  $|Q| \geq r[2] \sin(u/2) |\sqrt{1-\beta^2-\delta^2/3}|_+$ , useless as  $u \rightarrow 0$ , so short doubly critical pairs must be excluded otherwise. Version 0.2 did so with the curvature bound (a doubly critical arc of  $K_d$  has length  $\geq \pi/\kappa_d$ ), which under  $(H_3)$  gave only  $c_0 = 0.32, 0.13, 0.16$  at  $f = 1/2$ ,  $d = 1, 2, 3$ . The following, obtained during the second round of review, measures instead the turning of the tangent of  $K_d$  per unit base arclength and, for  $d \geq 2$ , uses no derivative of the base; it supersedes the v0.2 proposition, which is omitted.

**Lemma 3.9** (turning exclusion). *[proved here] Let  $\gamma$  be a  $C^1$  arc from  $X_1$  to  $X_2$  with unit tangent  $T_\gamma$ ,  $T_i$  its tangent at  $X_i$ ,  $a(\sigma) = \angle(T_\gamma(\sigma), T_1)$  and  $b(\sigma) = \angle(T_\gamma(\sigma), T_2)$ . If  $a + b < \pi$  along  $\gamma$ , then  $(X_2 - X_1) \cdot (T_1 + T_2) > 0$ ; in particular  $(X_1, X_2)$  is not doubly critical. This holds whenever the total turning  $\int_\gamma |T_\gamma'|$  is  $< \pi$ .*

*Proof.* As  $a, b \geq 0$  and  $a + b < \pi$ ,  $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2} > 0$ , so  $(X_2 - X_1) \cdot (T_1 + T_2) = \int_\gamma T_\gamma \cdot (T_1 + T_2) d\sigma > 0$ , while a doubly critical pair has  $(X_2 - X_1) \cdot T_1 = (X_2 - X_1) \cdot T_2 = 0$ . The last sentence follows from  $a(\sigma) \leq \int_0^\sigma |T_\gamma'|$  and  $b(\sigma) \leq \int_\sigma^\ell |T_\gamma'|$ .  $\square$

In the notation of the proof of Proposition 3.8,  $K_d' = v(1-x)T + rmW$  with  $x = r\kappa_\perp \in [-f, f]$ , so the angle  $\chi \in (0, \pi/2)$  between  $K_d'$  and  $T$  has  $\tan \chi = rm/(v(1-x))$  and  $\chi_{lo} := \arctan \frac{rm}{v_{\max}(1+f)} \leq \chi \leq \chi_{hi} := \arctan \frac{rm}{v_{\min}(1-f)} = \arctan \frac{1}{\zeta_{\min}}$ .

**Proposition 3.10** (pairs on the same strand,  $p = 2$ , sharpened). *[proved here] Let  $p = 2$ ,  $q$  odd,  $f = r/\tau \leq \frac{1}{2}$ ,  $m > 0$ . Set  $\delta_{R2} := (\pi - 2(\chi_{hi} - \chi_{lo})) / (1 + \sin \chi_{hi}/\Theta_{\min})$ ; if  $(H_3)$  of Proposition 3.8 holds, set  $\delta_{R1} := \pi/(\tau\bar{\Omega})$  with*

$$\tau\bar{\Omega} := 1 + \sin^2 \chi_{hi} + \frac{\sin \chi_{hi}}{\Theta_{\min}} + \tau \frac{rm[M_v(1+f)/v_{\min} + rM_\kappa]}{v_{\min}^2(1-f)^2 + r^2 m^2},$$

*and otherwise  $\delta_{R1} := 0$ ; let  $\delta_{\text{turn}} = \max(\delta_{R1}, \delta_{R2})$ . Then every doubly critical pair of  $K_d$  with distinct base points and strand index  $k = 0$  relative to the shorter base arc has  $\delta \geq \delta_{\text{turn}}$  and*

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c_0' r), \quad c_0' := \frac{1}{2r} \inf_{\delta_{\text{turn}} \leq \delta < \pi} \max(\Lambda(\delta), \ell_0(\delta) - 2r).$$

*In particular  $c_0' \geq \frac{1}{2}$  whenever  $\delta_{\text{turn}} \geq \pi/3$ .*

*Proof.* If  $\delta \geq \pi$  or the base chord is  $\geq 2\tau$ , Step 1 of Lemma 3.7 and Proposition 3.6(b) give  $|X_2 - X_1| \geq 2(\tau - r)$ . Otherwise let  $\gamma$  be the arc of  $K_d$  over the shorter base arc,  $t \in [t_1, t_2]$ , so  $u = m(t_2 - t_1) \leq \delta/\Theta_{\min}$ ; since  $k = 0$ ,  $\gamma$  joins  $X_1$  to  $X_2$ .

$\delta \geq \delta_{R2}$ , without  $(H_3)$ . Put  $\chi_i = \chi(t_i)$  and  $Y_i(t) = \cos \chi_i T(t) + \sin \chi_i W(t)$ . From  $U' = mW - v\kappa_\perp T$ ,  $W' = -mU - v\kappa_W T$ ,  $T' = v(\kappa_\perp U + \kappa_W W)$  we get  $Y_i' = (v\kappa_\perp \cos \chi_i - m \sin \chi_i)U + v\kappa_W \cos \chi_i W - v\kappa_W \sin \chi_i T$  and  $|Y_i'|^2 = (v\kappa_\perp \cos \chi_i - m \sin \chi_i)^2 + v^2 \kappa_W^2 \leq (v\kappa + m \sin \chi_i)^2$ , using  $\kappa_\perp^2 \cos^2 \chi_i + \kappa_W^2 \leq \kappa^2$  and  $-\kappa_\perp \cos \chi_i \leq \kappa$ . The unit tangent  $T_d(t) = \cos \chi(t) T(t) + \sin \chi(t) W(t)$  and  $Y_i(t)$  lie in  $\text{span}(T(t), W(t))$  at angles  $\chi(t)$ ,  $\chi_i$  from  $T(t)$ , and  $Y_i(t_i) = T_d(t_i)$ , so  $a(t) \leq |\chi(t) - \chi_1| + \int_{t_1}^t (v\kappa + m \sin \chi_1)$  and

$b(t) \leq |\chi(t) - \chi_2| + \int_t^{t_2} (v\kappa + m \sin \chi_2) dt$ . With  $\int v\kappa dt \leq \sigma/\tau = \delta$ ,  $a + b \leq 2(\chi_{\text{hi}} - \chi_{\text{lo}}) + \delta + u \sin \chi_{\text{hi}} \leq 2(\chi_{\text{hi}} - \chi_{\text{lo}}) + \delta(1 + \sin \chi_{\text{hi}}/\Theta_{\min})$ , which is  $< \pi$  if  $\delta < \delta_{R2}$ ; Lemma 3.9 excludes the pair.

$\delta \geq \delta_{R1}$ , under  $(H_3)$ . The turning rate of  $T_d$  per unit  $t$  is  $\omega = |K'_d \times K''_d|/|K'_d|^2$ . The Minkowski split in the proof of Proposition 3.8, multiplied by  $|K'_d| = \sqrt{A+B}$ , gives  $\omega \leq v\kappa(A+2B)/(A+B) + rm^2/\sqrt{A+B} + rm|a_0|/(A+B)$ , where  $B/(A+B) = \sin^2 \chi$  and  $rm/\sqrt{A+B} = \sin \chi$ ; with  $\kappa \leq 1/\tau$ ,  $\chi \leq \chi_{\text{hi}}$ ,  $v \geq v_{\min}$  and  $|a_0| \leq M_v(1+f) + rvM_\kappa$  this is  $\omega/v \leq \bar{\Omega}$  per unit base arclength, so the total turning of  $\gamma$  is  $\leq \tau\bar{\Omega}\delta$ , which is  $< \pi$  if  $\delta < \delta_{R1}$ .

*Distance.* For  $\delta < \pi$ , Step 3 of Lemma 3.7, which does not use the strand index, gives  $|X_2 - X_1| \geq \Lambda(\delta)$  and  $|X_2 - X_1| \geq \ell_0(\delta) - 2r$ ; hence  $c'_0$ . Finally  $\Lambda \geq r$  on  $[\pi/3, \pi)$  when  $f \leq \frac{1}{2}$ : on  $[\pi/3, \pi/2]$ ,  $\Lambda = 2(\tau - r) \sin(\delta/2) \geq \tau - r \geq r$ ; on  $[\pi/2, \pi)$  this is Step 4 of Lemma 3.7, and  $\Lambda/r$  decreases in  $f$ .  $\square$

**Corollary 3.11** (same strand in the default family). *[proved here] Let  $(p_j, q_j) = (2, 3)$  and  $f_j = r_j/\text{thick}(K_{j-1}) \leq \frac{1}{2}$  for all  $j \leq d$ . Then every doubly critical pair of  $K_d$  with distinct base points on the same strand is at distance  $\geq \min(2(\tau - r), 2c'_0 r)$  with  $c'_0 \geq \frac{1}{2}$ , at every depth  $d \geq 1$ , without  $(H_3)$ . At  $d = 1$  and  $f = \frac{1}{2}$ ,  $\delta_{\text{turn}} \geq \delta_{R1} = \pi/(1 + \frac{9}{13} + \frac{9}{2\sqrt{13}}) > 1.0684$  and  $c'_0 \geq \sin(\delta_{R1}/2) > 0.509$  (larger for smaller  $f$ ); for  $d \geq 2$ ,  $\delta_{\text{turn}} \geq \delta_{R2} > 1.677$ .*

*Proof.*  $d = 1$ . The base is the unit circle:  $v \equiv 1$ ,  $\kappa \equiv 1 = 1/\tau$ ,  $\alpha = 0$ ,  $m = \frac{3}{2}$ ,  $\Theta_{\min} = \frac{2}{3}$ , and  $(H_3)$  holds with  $M_v = M_\kappa = 0$ . With  $\zeta = \zeta_{\min} = 2(1-f)/(3f)$ ,  $\tau\bar{\Omega} = 1 + \frac{1}{1+\zeta^2} + \frac{3}{2\sqrt{1+\zeta^2}}$ , which is  $< 2.9404 < 3$  at  $f = \frac{1}{2}$  ( $\zeta = \frac{2}{3}$ ) and decreases as  $f$  decreases; so  $\delta_{R1} > \pi/3$ . On  $[\delta_{R1}, \pi/2]$ ,  $\Lambda/r = 2(1/f - 1) \sin(\delta/2) \geq c_* := 2 \sin(\delta_{R1}/2)$ . On  $[\pi/2, \pi)$  the argument of Step 4 of Lemma 3.7 with 1 replaced by  $c_*$  goes through:  $\phi_{c_*}(\zeta) = 4\zeta^2 - c_*\zeta - 1 + \pi/2 - 2 \arcsin \zeta$  has the same second derivative,  $\phi'_{c_*}(1/\sqrt{2}) = 2\sqrt{2} - c_* > 1.81$ ,  $\phi_{c_*}(1/\sqrt{2}) = 1 - c_*/\sqrt{2} > 0.279$  and  $\phi_{c_*}(1) = 3 - c_* - \pi/2 > 0.410$ . Both  $\delta_{R1}$  and  $\Lambda/r$  increase as  $f$  decreases.

$d \geq 2$ . By (5),  $|K'_1|^2 = (1-x)^2 + r_1^2 m_1^2$  with  $r_1 = f_1$ ,  $m_1 = \frac{3}{2}$ , so after the reparametrisation  $v_{\min}(K_1) \geq 2\sqrt{(1-f_1)^2 + \frac{9}{4}f_1^2}$ , while  $\text{thick}(K_1) \leq f_1$  (Proposition 3.5); hence  $v_{\min}(K_1)/\text{thick}(K_1) \geq 2\sqrt{(1-f_1)^2/f_1^2 + \frac{9}{4}} \geq \sqrt{13}$ . For  $j \geq 2$ ,  $v_{\min}(K_j) \geq 2(1-f_j)v_{\min}(K_{j-1})$  and  $\text{thick}(K_j) \leq f_j \text{thick}(K_{j-1})$ , so  $v_{\min}/\text{thick}$  at least doubles per depth. With  $m_d \leq \frac{3}{2} + \frac{1}{2} = 2$ ,  $\Theta_{\min} \geq \sqrt{13}2^{d-3} \geq \sqrt{13}/2 > 1.802$  and  $\zeta_{\min} = (1-f_d)\Theta_{\min}/f_d \geq \Theta_{\min}$ . Dropping  $\chi_{\text{lo}} \geq 0$ ,  $\delta_{R2} \geq (\pi - 2 \arctan(1/\zeta_{\min})) / (1 + 1/(\Theta_{\min}\sqrt{1+\zeta_{\min}^2}))$ , which increases in  $\Theta_{\min}$  and  $\zeta_{\min}$  and at  $\Theta_{\min} = \zeta_{\min} = \sqrt{13}/2$  exceeds  $1.677 > \pi/3$ . (The bound  $\Theta_{\min} \geq 2^{d-2}$  of Step 5 of Lemma 3.7 would not suffice at  $d = 2$ : at  $f = \frac{1}{2}$ ,  $\delta_{R2} \geq \pi/3$  needs  $\Theta_{\min} \geq 1.12$ .) Proposition 3.10 concludes.  $\square$

**Proposition 3.12** (curvature of  $K_d$  with the strand angle kept). *[proved here, under  $(H_3)$ ] Under the hypotheses of Proposition 3.8,  $\min\text{Rad}(K_d) \geq 1/\hat{\kappa}_d$ , where, with  $A = v^2(1-x)^2$ ,  $B = r^2m^2$  and  $b_U = v^2(1-x)x/r - rm^2$ ,*

$$\hat{\kappa}_d := \sup_{|x| \leq f, v_{\min} \leq v \leq v_{\max}} \frac{[b_U^2(A+B) + (rm(M_v(1-x) + rvM_\kappa) + v(A+2B)\sqrt{\tau^{-2} - x^2/r^2})^2]^{1/2}}{(A+B)^{3/2}}.$$

*Proof.* By the identity in the proof of Proposition 3.8,  $\kappa_{K_d}^2 = [b_U^2(A+B) + (rma_0 - \kappa_W v(A+2B))^2]/(A+B)^3$  with  $x = r\kappa_\perp$ ,  $|a_0| \leq M_v(1-x) + rvM_\kappa$  and  $|\kappa_W| = (\kappa^2 - \kappa_\perp^2)^{1/2} \leq (\tau^{-2} - x^2/r^2)^{1/2}$ ; for fixed  $(x, v)$  the right-hand side is non-decreasing in  $|a_0|$  and  $|\kappa_W|$ .  $\square$

**Theorem 3.13** (thickness lower bound for  $p = 2$ ). *[proved here; (b) under  $(H_3)$ ; the values in Table 1 are grid evaluations, not certified] Let  $p = 2$ ,  $q$  odd,  $f = r/\tau \leq 1/2$ ,  $m > 0$ . (a)  $\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, \min(c_1, c'_0)r)$ , with  $c_1 = c_1(f, \Theta_{\min})$  of Lemma 3.7 and  $c'_0$  of Proposition 3.10. (b) Under  $(H_3)$ ,  $\min\text{Rad}(K_d) \geq 1/\hat{\kappa}_d$ , hence  $\text{thick}(K_d) \geq \min(\tau - r, cr)$  with  $c = \min(c_1, c'_0, 1/(r\hat{\kappa}_d))$ . (c) For the default family  $(2, 3)$  with  $f_j \leq \frac{1}{2}$  at every level,  $\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, r/2) = r/2$  and  $\text{thick}(K_d) \geq \min(\min\text{Rad}(K_d), r/2)$  at every depth, without  $(H_3)$ ; at  $d = 1$ ,  $\text{thick}(K_1) \geq r_1/2$ .*

*Proof.* By (1),  $\text{thick} = \min(\min\text{Rad}, \frac{1}{2} \text{dcsd})$ . Doubly critical pairs with the same base point are at distance  $2r$  (Proposition 3.6(a)); if the shorter base arc has  $\delta \geq \pi$ , the base chord is  $\geq 2\tau$  (Step 1 of Lemma 3.7) and the distance is  $\geq 2(\tau - r)$  (Proposition 3.6(b)); pairs with  $\delta < \pi$  are covered by Lemma 3.7 ( $k = 1$ ) and Proposition 3.10 ( $k = 0$ ). Since  $c_1 \leq 1$  (its limit as  $\delta \rightarrow 0$  is 1), this gives (a); (b) is Proposition 3.12. For (c),  $c_1 \geq \frac{1}{2}$  is Lemma 3.7,  $c'_0 \geq \frac{1}{2}$  is Corollary 3.11, and  $\tau - r \geq r$ . At  $d = 1$  ( $M_v = M_\kappa = 0$ ,  $v \equiv 1$ ), Proposition 3.8 gives  $r\hat{\kappa}_1 = fG(\zeta_{\min})/(1-f) + 1/(1+\zeta_{\min}^2)$  with

$\zeta_{\min} = 2(1-f)/(3f)$  and  $G \leq \max g = g(\sqrt{2}) = 4\sqrt{2}/(3\sqrt{3})$  (the derivative of  $y(y+2)^2/(y+1)^3$ ,  $y = \zeta^2$ , is proportional to  $(y+2)(2-y)$ ); this increases with  $f$ , and at  $f = \frac{1}{2}$  equals  $g(\sqrt{2}) + \frac{9}{13}$ , so  $1/(r\bar{\kappa}_1) > 0.561 > \frac{1}{2}$ .  $\square$

Without the refinements of this version one has at  $f = 1/2$ ,  $d = 1, 2, 3$  only  $c_0 = 0.32, 0.13, 0.16$  and  $1/(r\bar{\kappa}_d) = 0.56, 0.48, 0.85$ . In Table 1,  $c \geq 0.509$  in every measured level, and the minimum is the different-strand constant  $c_1$  except at  $(f = \frac{1}{2}, d = 1)$ , where it is  $c'_0$ .

**Table 1:** Constants of Theorem 3.13 on the polygons ( $N_0 = 256$ ;  $\tau, v, \alpha$  and, for  $1/(r\hat{\kappa}_d)$  at  $d \geq 2$ ,  $M_v, M_\kappa$  measured on the base polygon;  $\delta_{\text{turn}} = \delta_{R1}$  at  $d = 1$ ,  $\delta_{R2}$  at  $d \geq 2$ ), rounded down. Grid evaluations, not certified (`experiments/check_Hc.py`, `check_same_strand.py`); analytic:  $c_1, c'_0 \geq \frac{1}{2}$  in every row. At  $f = 0.25$  all three are  $\geq 0.999$ . Last column: measured  $\tau_d/r_d$ .

| $f$  | $d$ | $c_1$ | $c'_0$ ( $\delta_{\text{turn}}$ ) | $1/(r\hat{\kappa}_d)$ | $c$   | $\tau_d/r_d$ |
|------|-----|-------|-----------------------------------|-----------------------|-------|--------------|
| 0.35 | 1   | 0.965 | 1.156 (1.344)                     | 2.215                 | 0.965 | 1.000        |
| 0.35 | 2   | 0.999 | 1.824 (2.838)                     | 1.878                 | 0.999 | 1.000        |
| 0.35 | 3   | 0.999 | 1.856 (3.107)                     | 1.855                 | 0.999 | 1.000        |
| 0.5  | 1   | 0.611 | 0.509 (1.068)                     | 1.419                 | 0.509 | 0.832        |
| 0.5  | 2   | 0.699 | 0.845 (2.056)                     | 1.034                 | 0.699 | 1.000        |
| 0.5  | 3   | 0.706 | 0.988 (2.928)                     | 0.998                 | 0.706 | 1.000        |

*Remark 3.14* (no constant works for an arbitrary base). [proved here] No  $c > 0$  depending only on  $(p, q, f)$  makes  $\text{thick}(K_d) \geq \min(\tau - r, cr \sin(\pi/p))$  true for every smooth base  $K$  with  $\text{thick}(K) = \tau$ ; the hypothesis of v0.1, stated for an arbitrary  $K$ , was therefore false (two counterexamples found in the first review). (i) If the curvature of  $K$  rises from 0 to  $1/\tau$  over an arc of length  $\varepsilon$  (a smoothed junction of a segment and a circle of radius  $\tau$ ), by (5) the tangent of the inner strand turns from  $vT + rmW$  to  $v(1-f)T + rmW$  over that arc, so  $\kappa_{K_d} \gtrsim rv\kappa'/|K'_d|^2 \rightarrow \infty$ : the  $M_v, M_\kappa$  term of (9) is unavoidable. (ii) If  $K$  is traversed slowly ( $v \rightarrow 0$ ) on an arc of length  $\gg r$ , the two strands there are helices of pitch  $h = v/m \rightarrow 0$ ; in the straight-tube model  $|K_d(t) - K_d(t + 2\pi + u)|^2 = 4r^2 \cos^2(qu/4) + v^2 u^2$ , whose minimum over  $u$  is  $\leq (2\pi v/q)^2$  and is attained at a doubly critical pair by screw symmetry, so  $\text{thick}(K_d) \leq \pi v/q \rightarrow 0$  (in Lemma 3.7,  $c_1 \rightarrow 0$  as  $\Theta_{\min} \rightarrow 0$ ). In the family  $\Theta_{\min}$  is bounded below a priori (Lemma 3.7, Corollary 3.11); no a priori bound on  $M_v, M_\kappa$  uniform in  $d$  is known.

What remains unproved is therefore a statement about the family, not about an arbitrary curve:

**Hypothesis 3.15** ( $H_c$ , for the family). [for  $(2, 3)$  and  $c = 1/2$ : proved at  $d = 1$ , reduced to a curvature bound for  $d \geq 2$  (Theorem 3.13); verified numerically] Let  $(K_d)$  be the family of Definition 2.1 built from the circle with a fixed pattern  $(p, q)$ , and let  $f \leq 1/2$ . There is  $c = c(p, q, f) \in (0, 1]$  such that, for every sequence  $(r_d)$  with  $r_d \leq f \text{thick}(K_{d-1})$  and every  $d \geq 1$ ,

$$\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, c r_d \sin(\pi/p)). \quad (10)$$

*Status for  $(2, 3)$ ,  $f \leq 1/2$ ,  $c = 1/2$ .* By Theorem 3.13(c), (10) holds at  $d = 1$  and, for  $d \geq 2$ , is equivalent to  $\min\text{Rad}(K_d) \geq r_d/2$ . Under  $(H_3)$  this follows from  $1/(r\hat{\kappa}_d) \geq 1/2$ , which the grid evaluation with polygon-measured inputs gives at  $d = 2, 3$  (1.034, 0.998 at  $f = 1/2$ ; Table 1); this is evidence, not a proof, since neither  $\text{thick}(K_{d-1})$  nor  $M_v, M_\kappa$  is known a priori. Open: an a priori bound for  $(H_3)$  uniform in  $d$ , and certification. Numerically (10) with  $c = 1/2$  holds in all 9 default levels with margin  $\tau_d/\rho_d \geq 1.66$ , and along the chain of Corollary 3.16 with margin  $\geq 1.66$ .

**On the constant.** The natural constant is  $c = 1$ :  $r \sin(\pi/p)$  is half the strand chord, and by Propositions 3.5 and 3.6(a) it cannot be improved where  $\tau_d = r_d$ . With  $c = 1$  the inequality fails on the polygons in 4 of the 15 measured levels with  $f \leq 1/2$  ( $(2, 3)$ ,  $f = 0.5$ ,  $d = 1$  (0.832);  $(3, 2)$ ,  $d = 1$  (0.968);  $(3, 2)$ ,  $d = 2$  (0.999);  $(2, 5)$ ,  $d = 1$  (0.564); the deficit at  $(3, 2)$ ,  $d = 2$  is stable under doubling of  $N$ ); it holds with equality, to 0.05%, in every level with  $\tau_d = r_d$  and in  $(3, 2)$ ,  $d = 3$ ;  $(2, 5)$ ,  $d = 2, 3$ , where the critical pair is a pair of adjacent strands of one disc. The value  $c = 1/2$  was chosen after  $c = 1$  had failed (Section 4); Theorem 3.13 reaches it, while  $\tau_1/r_1 = 0.832$  at  $f = 1/2$  shows that no argument reaches  $c = 1$  there.

**Corollary 3.16** (recursion for the ropelength bound). [proved here, conditional on  $H_c$  for the family (for  $(2, 3)$ ,  $c = 1/2$ : on  $\min\text{Rad}(K_d) \geq r_d/2$ ,  $d \geq 2$ )] Fix  $(p, q)$ ,  $f \leq 1/2$  and assume  $(H_c)$  with constant  $c$  (for all  $r_d \leq f \text{thick}(K_{d-1})$ ). Define  $\rho_0 = 1$ ,  $r_d := f\rho_{d-1}$  and  $\rho_d := \gamma\rho_{d-1}$  with

$$\gamma = \min(1 - f, cf \sin(\pi/p)), \quad A = p(1 + f) + f(q + \frac{p}{2}). \quad (11)$$

Then  $\text{thick}(K_d) \geq \rho_d = \gamma^d$  and  $L(K_d) \leq 2\pi A^d$  for all  $d$ , hence  $\text{Rop}(K_d) \leq 2\pi\Lambda^d$  with  $\Lambda = A/\gamma$ . For  $(2, 3)$  and  $f = 1/2$ :  $A = 5.00$ , and  $\Lambda = 20.0$  with  $c = 1/2$  ( $\gamma = 0.250$ ),  $\Lambda = 10.0$  with  $c = 1$  ( $\gamma = 0.500$ ).

*Proof.* Induction on  $d$ . If  $\text{thick}(K_{d-1}) \geq \rho_{d-1}$  then  $r_d = f\rho_{d-1} \leq f\text{thick}(K_{d-1})$ , so  $(H_c)$  applies and  $\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, cr_d \sin(\pi/p)) \geq \min(\rho_{d-1} - f\rho_{d-1}, cf\rho_{d-1} \sin(\pi/p)) = \gamma\rho_{d-1} = \rho_d$ . By Lemma 3.3 with  $\tau \geq \rho_{d-1}$ ,  $L(K_d) \leq pL(K_{d-1})(1+f) + f\rho_{d-1}(2\pi q + \pi p)$ ; since  $\rho_{d-1} \leq \text{thick}(K_{d-1}) \leq L(K_{d-1})/2\pi$ , this is  $\leq L(K_{d-1})[p(1+f) + f(q+p/2)] = AL(K_{d-1})$ , and  $L(K_0) = 2\pi$ .  $\square$

*Remark 3.17* (on the constants).  $\Lambda$  is generous: Lemma 3.3 bounds  $|K'_d|$  by the sum of the norms of two orthogonal vectors, uses  $\kappa_{\max} \leq 1/\tau$  and  $|\alpha| \leq \pi$ , and  $(H_c)$  with  $c = 1/2$  is twice weaker than the observed  $\tau_d \approx r_d$ . Numerically the growth factor per depth is 4.06, 4.00 at  $f = 1/2$  (Table 2) against  $\Lambda = 20.0$ ; along the chain of the corollary itself ( $r_d = f\rho_{d-1}$ , `experiments/corollary_chain.py`) the polygons give  $L/\tau = 38.4, 256.5, 2051.9$  against  $2\pi\Lambda^d = 126, 2513, 50265$ , and  $\tau_d/\rho_d \geq 1.66$  for  $f \in \{0.25, 0.35, 0.5\}$ . We make no claim that any configuration in the family is stationary or optimal for ropelength, and none that the recursion is a “flow law” for tight recursive knots: it bounds one explicit construction.

*Remark 3.18* (lower bounds, quoted). [literature] Buck and Simon [2] prove  $\text{Rop}(K) \geq c_{\text{BS}} \text{Cr}(K)^{3/4}$  for a universal  $c_{\text{BS}} > 0$ , and Cantarella, Kusner and Sullivan [3] prove that ropelength minimisers exist in every knot and link type, are  $C^{1,1}$ , and satisfy further lower bounds. These need  $\text{Cr}(K_d)$ .  $K_2$  is a 2-cable of the trefoil (Lemma 3.2), which is alternating, hence adequate; Kalfagianni and McConkey [8] prove that a cable with  $q$  strands of an adequate knot  $K$  has crossing number  $\geq q^2 \text{Cr}(K) + 1$ , so  $\text{Cr}(K_2) \geq 13$ , and they determine the crossing numbers of 2-cables of adequate knots (we have not matched their cable and writhe conventions with Lemma 3.2, so we do not quote the exact value). For  $d \geq 3$  only general satellite bounds are available, such as  $\text{Cr} \geq 10^{-13} \text{Cr}(\text{companion})$  [9]. For  $K_1 = T(2, 3)$  the best known numerical ropelength is about 32.7 (numerical, uncertified; [3, 12]); our depth-one polygon at  $f = 1/2$  gives 38.4, i.e. the construction is not close to tight even at depth one.

**Conjecture 3.19** (a threshold for  $\text{thick}(K_d) = r_d$ ). [conjectural] For the default family there are thresholds  $f_*(d) > 0$  with  $\inf_d f_*(d) > 0$  such that for  $f \leq f_*(d)$  every doubly critical pair of  $K_d$  other than the antipodal pairs of one normal disc is at distance  $> 2r_d$  and  $\min\text{Rad}(K_d) > r_d$ ; hence  $\text{thick}(K_d) = r_d$  (the inequality  $\leq$  is Proposition 3.5). Numerically the largest admissible  $f_*(1)$  lies in  $[0.35, 0.5]$ , and  $f_*(2), f_*(3) \geq 0.5$ , the latter by margins of only 1.4% and 2.1% (Section 4).

*Remark 3.20* (growth rate). If  $\text{thick}(K_d) = r_d$  for all large  $d$  and  $\min\text{Rad}(K_{d-1})$  stays bounded below, then  $\text{Rop}_d/\text{Rop}_{d-1} = (L_d/L_{d-1})(\tau_{d-1}/\tau_d) \rightarrow p/f$  by Lemma 3.3. The measured ratios at  $d = 2, 3$  are 8.01, 8.00 ( $f = 0.25$ ;  $p/f = 8.00$ ), 5.73, 5.71 (5.71), 4.06, 4.00 (4.00), 4.03, 4.00 ((2, 5); 4.00) and 6.95, 6.93 ((3, 2), adjacent strands, limit  $p/(f \sin(\pi/p)) = 6.93$ ): the rate is a consequence of the conjecture, not an independent constant.

## 4 Experiments

**Criteria.** The protocol is in Appendix A. Four criteria are checked: (C1) resolution:  $\text{Rop}$  and  $\tau$  change by less than 1% between  $N_0$  and  $2N_0$ ; (C2)  $L(K_d) \leq$  the right-hand side of (6) evaluated with the measured  $L(K_{d-1})$ ,  $\tau(K_{d-1})$  and  $\alpha_d$ ; (C3)  $\tau(K_d) \geq \rho_d$  of (10) with  $c = 1/2$  and the measured  $\tau(K_{d-1})$ , for  $f \leq 1/2$ ; (C4)  $\tau_{pt}$  agrees with  $\tau$  to within 1%. The plan written before any code (`PLAN.md`) fixed the resolution check at  $d = 1, 2$  without a threshold, the length bound and the form of (10) with an unspecified constant; the 1% thresholds, (C1) at  $d = 3$  and (C4) were fixed when the manuscript was written, and  $c = 1/2$  in (C3) after a first run with  $c = 1$  had failed at  $(f = 1/2, d = 1)$ : (C3) is a post hoc check.

**Results.** Table 2 gives all patterns at the finest resolution and Figure 1 the growth (projections of the curves: `figures/fig_curves.pdf`). All four criteria hold: (C1) the maximal relative change is 0.03% for  $\text{Rop}$  and 0.03% for  $\tau$  over all 12 chains (per-level values in `results/tables.md`); (C2) holds in 36 of 36 levels; (C3) holds in all 9 default levels with  $f \leq 1/2$ , with margin  $\tau_d/\rho_d \geq 1.66$  (and  $\geq 1.13$  including the other patterns;  $c = 1$ : see after Hypothesis 3.15); (C4) the maximal discrepancy is 0.01%. In every level  $\tau$  is determined by  $\text{dcsd}/2$ , never by  $\min\text{Rad}$  (36 of 36;  $\min\text{Rad}/r_d \geq 1.15$  throughout).

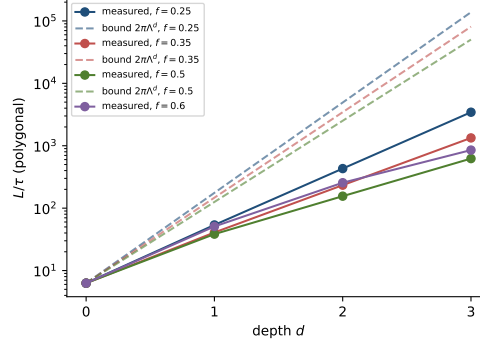
Three further observations. First, for  $f \leq 0.35$  the measured  $\tau(K_d)$  equals  $r_d$  to four digits at every depth (the antipodal strands in a normal disc are the doubly critical pair, consistent with Proposition 3.5 and Conjecture 3.19), so  $\text{Rop}_d = L_d/r_d$  and the recursion  $\tau_d = f\tau_{d-1}$  is exact; for  $f = 1/2$  at depth one  $\tau_1 = 0.4158 < r_1 = 0.5$ , realised by the pair across the hole of the torus described in Section 3,

while at depths 2, 3 again  $\tau_d = r_d$ , but narrowly: the next-closest doubly critical vertex pairs (base points at arc  $\geq \pi\tau$ ) are only 1.4% and 2.1% farther than  $2r_d$  (1.4% and 2.0% at  $N_0 = 512$ ; 103% and 109% at  $f = 0.35$ ), so  $f = 1/2$  is close to the threshold of Conjecture 3.19. In the same census (`experiments/classify_pairs.py`, 18 levels of the default family,  $f \leq 1/2$ ,  $N_0 \in \{256, 512\}$ ) no doubly critical vertex pair on the same strand with distinct base points occurs; a sign-change test that also detects saddles and maxima of the distance finds same-strand doubly critical pairs only at  $d = 1$ , with  $\delta \geq 1.96 > \delta_{\text{turn}}$  and distance  $\geq 3.70 r_1$  at  $f = 1/2$ , and none at  $d = 2, 3$ . Second, the writhe of the depth-one configuration crosses the half-integer 3.5 inside the grid ( $\text{Wr}(K_1) = 3.127, 3.257, 3.518, 3.710$  for  $f = 0.25, 0.35, 0.5, 0.6$ ; the crossing is at  $r_1^* = 0.4904$  by bisection), so by Lemma 3.2 the frame linking number is 3 for  $f \leq 0.35$  and 4 for  $f \geq 0.5$  and the depth-two knot type changes across the grid:  $K_2$  is the  $(2, 9)$ -cable of the trefoil for  $f = 0.25$  but the  $(2, 11)$ -cable for  $f = 0.5$ . The residual  $|\text{Wr } K_{d-1} - \alpha_d/2\pi - n|$ , a consistency check of Lemma 3.2, is at most  $1.1 \cdot 10^{-3}$  at the fine resolution ( $4.5 \cdot 10^{-3}$  at the coarse one). The midpoint-rule writhe changes under doubling of  $N$  by at most  $1 \cdot 10^{-4}$  at depth one of the default family but by up to 0.031 at depth three of  $(3, 2)$ ; the knot-type decisions are unaffected (smallest margin to a half-integer 0.018, at  $f = 1/2$ ,  $d = 1$ ). Third,  $f = 0.5$  gives the best value on the grid at depths one and three;  $f = 0.6$  is worse at every depth;  $f$  was not optimised.

**Curve-shortening probe (uncertified).** 200 explicit discrete curve-shortening steps ( $\lambda = 0.1$ , `experiments/shrink.py`) lower  $L/\tau$  of  $K_1$  ( $f = 1/2$ ,  $N_0 = 512$ ) from 38.4 to 38.1 and raise that of  $K_2$  from 155.6 to 156.3: no evidence about local minimality (`results/tables_shrink.md`).

**Table 2:** All chains at the finest resolution ( $N_0 = 1024$ ; 512 for  $(3, 2)$ ).  $r_d = f\tau_{d-1}$ ;  $\rho_d^{(1)} = \min(\tau_{d-1} - r_d, r_d \sin(\pi/p))$  is (10) with the natural constant  $c = 1$ , so  $\tau_d/\rho_d^{(1)} < 1$  marks a failure of  $c = 1$  (the ratio for  $c = 1/2$  is twice as large when  $\tau_{d-1} - r_d$  is not the minimum); slope =  $q + pn$  of Lemma 3.2.  $^\dagger$ :  $f = 0.6$  lies outside the hypothesis  $r \leq \tau/2$  and is a numerical probe only. Lengths in units of  $R_0$ .

| $(p, q)$ | $f$            | $d$ | $N$   | $r_d$  | $L$     | $\tau$ | $L/\tau$ | $\tau_d/\rho_d^{(1)}$ | Wr     | slope |
|----------|----------------|-----|-------|--------|---------|--------|----------|-----------------------|--------|-------|
| (2, 3)   | 0.25           | 1   | 2048  | 0.2500 | 13.444  | 0.2500 | 53.8     | 1.000                 | 3.13   | 3     |
| (2, 3)   | 0.25           | 2   | 4096  | 0.0625 | 26.911  | 0.0625 | 430.6    | 1.000                 | 15.25  | 9     |
| (2, 3)   | 0.25           | 3   | 8192  | 0.0156 | 53.822  | 0.0156 | 3444.6   | 1.000                 | 63.50  | 33    |
| (2, 3)   | 0.35           | 1   | 2048  | 0.3500 | 14.271  | 0.3500 | 40.8     | 1.000                 | 3.26   | 3     |
| (2, 3)   | 0.35           | 2   | 4096  | 0.1225 | 28.608  | 0.1225 | 233.5    | 1.000                 | 15.49  | 9     |
| (2, 3)   | 0.35           | 3   | 8192  | 0.0429 | 57.219  | 0.0429 | 1334.6   | 1.000                 | 63.98  | 33    |
| (2, 3)   | 0.5            | 1   | 2048  | 0.5000 | 15.949  | 0.4158 | 38.4     | 0.832                 | 3.52   | 3     |
| (2, 3)   | 0.5            | 2   | 4096  | 0.2079 | 32.349  | 0.2079 | 155.6    | 1.000                 | 17.99  | 11    |
| (2, 3)   | 0.5            | 3   | 8192  | 0.1040 | 64.730  | 0.1040 | 622.7    | 1.000                 | 74.96  | 39    |
| (2, 3)   | 0.6 $^\dagger$ | 1   | 2048  | 0.6000 | 17.307  | 0.3401 | 50.9     | 0.850                 | 3.71   | 3     |
| (2, 3)   | 0.6 $^\dagger$ | 2   | 4096  | 0.2041 | 34.941  | 0.1367 | 255.6    | 1.005                 | 18.36  | 11    |
| (2, 3)   | 0.6 $^\dagger$ | 3   | 8192  | 0.0820 | 69.893  | 0.0820 | 852.1    | 1.500                 | 75.70  | 39    |
| (3, 2)   | 0.5            | 1   | 1536  | 0.5000 | 20.000  | 0.4191 | 47.7     | 0.968                 | 3.90   | 2     |
| (3, 2)   | 0.5            | 2   | 4608  | 0.2096 | 60.087  | 0.1813 | 331.4    | 0.999                 | 39.66  | 14    |
| (3, 2)   | 0.5            | 3   | 13824 | 0.0906 | 180.270 | 0.0785 | 2296.5   | 1.000                 | 362.97 | 122   |
| (2, 5)   | 0.5            | 1   | 2048  | 0.5000 | 20.420  | 0.2822 | 72.4     | 0.564                 | 6.87   | 5     |
| (2, 5)   | 0.5            | 2   | 4096  | 0.1411 | 41.116  | 0.1411 | 291.4    | 1.000                 | 32.66  | 19    |
| (2, 5)   | 0.5            | 3   | 8192  | 0.0705 | 82.273  | 0.0705 | 1166.2   | 1.000                 | 136.30 | 71    |



**Figure 1:** Polygonal  $L/\tau$  against depth for the default family (solid) and the conditional bound  $2\pi\Lambda^d$  with  $c = 1/2$  (dashed,  $f \leq 1/2$ ).

## 5 Claims and their status

| Claim                                                                                                                                 | Status                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| $K_d$ embedded, smooth, regular if $r_d < \text{thick}(K_{d-1})$ (Lemma 3.1).                                                         | proved here                                            |
| $\text{Lk} = \text{Wr} - \alpha/2\pi$ ; $K_d$ is the $(p, q + pn)$ -cable of $K_{d-1}$ (Lemma 3.2).                                   | proved here from classical results                     |
| Two-sided length bounds (6) (Lemma 3.3).                                                                                              | proved here                                            |
| $p = 2$ : $\text{thick}(K_d) \leq r_d$ , $\text{Rop}(K_d) \geq L(K_d)/r_d \geq 2\pi(p(1-f)/f)^d$ (Prop. 3.5).                         | proved here (internal to the family)                   |
| Same-disc pairs $\geq 2r \sin(\pi/p)$ ; far pairs $\geq 2(\tau - r)$ (Prop. 3.6).                                                     | proved here                                            |
| Different strands: $\geq \min(2(\tau - r), 2c_1 r)$ , $c_1 \geq \frac{1}{2}$ in $(2, 3)$ , all $d$ (Lemma 3.7).                       | proved here; value 0.6117 is a grid value              |
| Same strand: $\geq \min(2(\tau - r), 2c'_0 r)$ , $c'_0 \geq \frac{1}{2}$ in $(2, 3)$ , all $d$ , no $(H_3)$ (Prop. 3.10, Cor. 3.11).  | proved here (round 2); refined values are grid values  |
| $\text{minRad}(K_d) \geq 1/\bar{\kappa}_d$ , $\geq 1/\hat{\kappa}_d$ (Props. 3.8, 3.12).                                              | proved under $(H_3)$ ; unconditional at $d = 1$        |
| $(2, 3)$ , $f \leq \frac{1}{2}$ : $\frac{1}{2} \text{dcsd}(K_d) \geq r_d/2$ for all $d$ ; $\text{thick}(K_1) \geq r_1/2$ (Thm. 3.13). | proved here                                            |
| $\text{thick}(K_d) \geq r_d/2$ at $d = 2, 3$ (Table 1).                                                                               | grid evaluation under $(H_3)$ , polygon inputs         |
| No $c > 0$ depending only on $(p, q, f)$ works for every base (Remark 3.14).                                                          | proved here                                            |
| $(H_c)$ , $(2, 3)$ , $c = \frac{1}{2}$ , all $d$ (needs $(H_3)$ uniformly in $d$ ).                                                   | open; numerically true in 9 levels, margin $\geq 1.66$ |
| $\text{Rop}(K_d) \leq 2\pi\Lambda^d$ , $\Lambda = A/\gamma$ (Cor. 3.16).                                                              | proved, conditional on $(H_c)$                         |
| Polygons with $L/\tau = 38.4, 155.6, 622.7$ ( $f = 0.5$ , $d = 1, 2, 3$ ), stable to 0.03%.                                           | computed (polygonal proxy, uncertified)                |
| $K_2$ changes knot type between $f = 0.35$ and $0.5$ ( $r_1^* = 0.4904$ ).                                                            | computed; consequence of Lemma 3.2                     |
| $\text{Rop} \geq c_{BS} \text{Cr}^{3/4}$ ; tight knots; $\text{Cr}(K_2) \geq 13$ ; satellites.                                        | literature [2, 3, 8, 9]                                |
| $\text{thick}(K_d) = r_d$ for $f \leq f_*(d)$ ; then $\text{Rop}_d / \text{Rop}_{d-1} \rightarrow p/f$ (Conj. 3.19).                  | conjectural; margins 1.4%, 2.1% at $f = \frac{1}{2}$   |
| Curve shortening: $L/\tau$ of $K_1$ $-0.8\%$ , of $K_2$ up.                                                                           | uncertified probe; negative at $d = 2$                 |
| Stationarity, optimality, a recursive flow law.                                                                                       | not claimed                                            |

## 6 Limitations and next steps

*Limitations.* (1) The thickness lower bound ( $p = 2$ , default family) is proved for all doubly critical pairs at every depth and completely at  $d = 1$ ; for  $d \geq 2$  the curvature of  $K_d$  needs  $(H_3)$ , whose constants are only measured on the polygons and whose uniformity in  $d$  is open. (2) The polygonal  $\tau$  of (4) (vertex pairs, local-minimum test) is supported by  $\tau_{pt}$ , the segment check and stability in  $N$ , but is not certified. (3) Nothing is optimised; the family (pattern angle linear in  $t$ ,  $\varphi_d = 0$ ) is one of many; for  $p \geq 3$  the thickness lower bound is treated only in Proposition 3.6. (4) Only the thresholds  $c_1 \geq \frac{1}{2}$ ,  $c'_0 \geq \frac{1}{2}$  and  $1/(r\bar{\kappa}_1) > 0.561$  are analytic; every other constant of Section 3 is a grid infimum or supremum ( $c_1$ ,  $c'_0$  with a Lipschitz constant estimated on the grid itself,  $\hat{\kappa}_d$  with a neighbour-increment correction, the v0.2 constant  $c_0$  quoted for comparison on a  $1500 \times 400$  grid without correction), with polygon-measured inputs for  $d \geq 2$ ; none uses interval arithmetic. (5) Writhe and holonomy are midpoint-rule sums, stable to  $1 \cdot 10^{-4}$  at depth one but only to 0.031 at depth three of  $(3, 2)$ ; the frame linking number for  $f = 0.5$  is decided by a margin of 0.018; an exact polygonal writhe is not implemented. (6) The only lower bound specific to the family is Proposition 3.5; knot-type bounds are limited by what is known of  $\text{Cr}(K_d)$  (Remark 3.18).

*Next steps.* (1) Bound  $M_v$ ,  $M_\kappa$  of  $K_d$  in terms of those of  $K_{d-1}$  to make  $(H_3)$  uniform in  $d$ ; with Theorem 3.13 this would prove  $(H_c)$  with  $c = 1/2$  for  $(2, 3)$  at every depth, and Corollary 3.16 unconditionally. (2) Certify: interval arithmetic for  $c_1$ ,  $c'_0$ ,  $\hat{\kappa}_d$  and for  $\text{minRad}$ ,  $\text{dcsd}$  of a piecewise-circular

interpolant; exact polygonal writhe (Levitt; Klenin–Langowski). (3) Replace the curve-shortening probe by a constrained length minimisation at fixed  $\tau \geq 1$  with a certified knot-type check, for  $d = 1, 2$ . (4) Optimise  $f_d$  per depth and the pattern angle in arclength; treat  $p \geq 3$  (the bound  $\delta_{R2}$  is general, the a priori bounds use  $m \leq 2$  and  $m_1 = 3/2$ ). (5) Match the conventions of [8] with Lemma 3.2 to obtain  $\text{Cr}(K_2)$  and attach a Buck–Simon bound to each depth-two configuration.

## A Reproducibility and protocol

*Protocol.* `experiments/recursive_knots.py` builds the polygon  $K_0$  with  $N_0$  vertices, computes the closed Bishop frame by discrete parallel transport (Rodrigues rotation taking  $T_i$  to  $T_{i+1}$ ; holonomy = signed angle between transported and initial normal, removed by the twist  $-\alpha i/N$ ) and produces  $K_d$  with  $p_d N_{d-1}$  vertices by (3) at the vertices ( $t$  is the vertex index). At each depth it measures  $L$ ,  $\text{minRad}$ ,  $\text{dcsd}$ ,  $\tau$ ,  $\tau_{pt}$ ,  $\text{Rop}$ , writhe, holonomy and  $\text{Lk} = \text{rint}(\text{Wr } K_{d-1} - \alpha_d/2\pi)$  with its residual, and sets  $r_{d+1} = f \tau(K_d)$ . Grid:  $(2, 3)$  with  $f \in \{0.25, 0.35, 0.5, 0.6\}$  ( $f = 0.6$  a probe outside  $r \leq \tau/2$ ),  $(3, 2)$ ,  $(2, 5)$  with  $f = 0.5$ ; depths 1–3;  $N_0 \in \{512, 1024\}$  ( $\{256, 512\}$  for  $(3, 2)$ ). Deterministic (seed 20260930 recorded only); 184s wall-clock on shared cores, Python 3.11.15, NumPy 2.4.6. `aux_checks.py`: critical pair at  $d = 1$ , writhe crossing, segment check. `check_Hc.py`, `check_same_strand.py`: constants of Section 3 and every intermediate inequality of the proofs on the polygons; frozen copies for v0.3 of the scripts in `theory/` where the derivations were done (paths changed only; outputs byte-identical; SHA-256 in `results/FROZEN_THEORY_SHA256.txt`). `classify_pairs.py`: census of doubly critical pairs; `corollary_chain.py`: chain of Corollary 3.16. `make_numbers.py` writes `manuscript/numbers.tex` and the table body from `results/*.json` only; `manuscript/build.sh` runs it and `latexmk`. Every number in this text is a macro from `numbers.tex`.

## References

- [1] Richard L. Bishop. There is more than one way to frame a curve. *American Mathematical Monthly*, 82(3): 246–251, 1975.
- [2] Gregory Buck and Jonathan Simon. Thickness and crossing number of knots. *Topology and its Applications*, 91(3):245–257, 1999.
- [3] Jason Cantarella, Robert B. Kusner, and John M. Sullivan. On the minimum ropelength of knots and links. *Inventiones Mathematicae*, 150(2):257–286, 2002. doi: 10.1007/s00222-002-0234-y.
- [4] Gheorghe Călugăreanu. Sur les classes d’isotopie des nœuds tridimensionnels et leurs invariants. *Czechoslovak Mathematical Journal*, 11(4):588–625, 1961. doi: 10.21136/CMJ.1961.100486.
- [5] Werner Fenchel. Über Krümmung und Windung geschlossener Raumkurven. *Mathematische Annalen*, 101: 238–252, 1929. doi: 10.1007/BF01454836.
- [6] F. Brock Fuller. The writhing number of a space curve. *Proceedings of the National Academy of Sciences of the USA*, 68(4):815–819, 1971. doi: 10.1073/pnas.68.4.815.
- [7] Oscar Gonzalez and John H. Maddocks. Global curvature, thickness, and the ideal shapes of knots. *Proceedings of the National Academy of Sciences of the USA*, 96(9):4769–4773, 1999.
- [8] Efstratia Kalfagianni and Rob McConkey. Crossing numbers of cable knots. *Bulletin of the London Mathematical Society*, 56(11):3400–3411, 2024. doi: 10.1112/blms.13140.
- [9] Marc Lackenby. The crossing number of satellite knots. *Algebraic & Geometric Topology*, 14(4):2379–2409, 2014. doi: 10.2140/agt.2014.14.2379.
- [10] W. B. Raymond Lickorish. *An Introduction to Knot Theory*, volume 175 of *Graduate Texts in Mathematics*. Springer, New York, 1997.
- [11] R. A. Litherland, J. Simon, O. Durumeric, and E. Rawdon. Thickness of knots. *Topology and its Applications*, 91(3):233–244, 1999.
- [12] Piotr Pieranski. In search of ideal knots. *Computational Methods in Science and Technology*, 4:9–23, 1998. doi: 10.12921/cmst.1998.04.01.09-23.
- [13] Eric J. Rawdon. Can computers discover ideal knots? *Experimental Mathematics*, 12(3):287–302, 2003.
- [14] James H. White. Self-linking and the Gauss integral in higher dimensions. *American Journal of Mathematics*, 91(3):693–728, 1969. doi: 10.2307/2373348.